

Cairo University - Faculty of Engineering

Computer Engineering Department

Big Data

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Big Data

Team 1

Project Proposal

Team Members:

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Ideas Proposed:

- Airbnb Global Listings

Problem: Airbnb Global Listings

Idea:

The short-term rental market has grown significantly over the past decade, with platforms like Airbnb disrupting the traditional hospitality industry. This project aims to analyze the key factors that determine Airbnb listing prices and identify what makes a listing successful. The insights will help hosts optimize their pricing strategy, improve occupancy rates, and help travelers find the best value listings.



Business Questions:

- What factors most influence the price of an Airbnb listing?
 - Which locations and property types generate the highest occupancy rates?
 - Can we predict listing price based on property features, location, and host characteristics?
 - What amenities and features are commonly associated with highly-rated listings?
 - How many listings are rented frequently to tourists vs. long-term residents?
 - Which hosts are running a business with multiple listings and where are they located
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Dataset:

- 89 features including property details, host information, pricing, availability, and reviews
- Data sourced from publicly available information from the Airbnb platform, analyzed and cleansed by Inside Airbnb
- Features include location coordinates, host characteristics, pricing, availability, and guest reviews

- All features are clearly labeled and interpretable
 - size:1.94GB
 - Link: <https://www.kaggle.com/datasets/joebeachcapital/airbnb>
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Planned Approach:

- **Preprocessing & EDA:** Clean the data by handling missing values, removing irrelevant columns such as URLs and free-text fields, encoding categorical features, and visualizing price distributions across cities, property types, and room types.
 - **Descriptive Analysis:** we'll apply K-Means clustering to segment listings by location and price range, and use Association Rules to find common amenity combinations in highly-rated listings.
 - **Predictive Analysis:** we'll train multiple classifiers including Logistic Regression, KNN, and Decision Tree to predict whether a listing is high or low demand, and apply Linear Regression to predict listing price.
 - **MapReduce:** we'll implement KNN using Hadoop/Spark in pseudo-distributed mode.
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